Now You See Me, Now You Don't: Consent-Driven Privacy for Smart Glasses

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***Abstract*—Smart glasses pose significant privacy challenges by capturing bystanders without notice or consent. Existing solutions often rely on permanent obfuscation, shift responsibility to bystanders, or offload sensitive data to the cloud, risking unauthorized access and denying meaningful control. We present a novel three-tier architecture for privacy-preserving smart glasses that enforces blurring at the point of capture, supports synthetic face replacement, and enables consent-based decryption of visual data. We implement SITARA, a working prototype on Raspberry Pi–based hardware, and demonstrate on-device blurring and secure consent mediation. Our evaluation shows that SITARA operates efficiently while achieving reliable bystander anonymization; furthermore, its synthetic replacement delivers perceptual quality competitive with state-of-the-art baselines, all without exposing raw video or undermining usability.**

***Index Terms*—smart glasses, privacy, bystander consent, wearable computing, computer vision**

I. Introduction

Smart glasses are rapidly moving from fiction to reality, with devices like Ray-Ban Meta [1], Meta's Orion AR [2], and Google's Android XR [3] entering everyday use. They also raise serious privacy risks by indiscriminately capturing individuals and enabling unwanted identification. Existing privacy indicators such as LED recording lights are unreliable, often ignored, misunderstood, or disabled [4], leaving most people unaware when recording occurs. Surveys show widespread discomfort and growing tension around smart eyewear [5], with bystanders feeling unprotected and lacking means to prevent unwanted data collection [6]. At the same time, recent research indicates that wearers also support stronger privacy safeguards and often feel the burden of protecting bystanders' privacy, underscoring the need for systems that relieve wearers of the responsibility of managing bystander privacy [7].

Existing Privacy-Enhancing Technologies (PETs) attempt to address these gaps but remain limited. Systems such as SnapMe, FaceBlock, SelfFlag, and I-Pic [8]–[11] require bystanders to take the initiative by broadcasting policies. This shifts the burden of privacy protection onto bystanders, who must install apps, wear devices, or perform gestures. These systems provide weak conditional privacy and lack mechanisms to restore visibility if consent is granted later. Cloud-based approaches such as EgoBlur [12] and Google Street View [13], rely on irreversible blurring and server-side processing. While protective, these methods offer no way to reverse obfuscation and depend on external servers, raising concerns about exposing sensitive data.

We argue that meaningful progress requires solutions that serve both wearers and bystanders by re-architecting data flow around three key principles: *privacy at the point of capture*, *recording usability* and *bystander-owned disclosure control*. We implement these principles through a three-tier architecture: ❶ a mandatory on-device blurring mechanism shields bystanders immediately, removing the burden of privacy protection; ❷ an optional layer replaces blurred faces with photorealistic, computer-generated substitutes providing suitable alternatives for wearers; and ❸ a consent-based restoration mechanism allows unblurring only with the bystander's cryptographic consent, enabling better consent negotiation and data control for both wearers and bystanders.

We validate the design with SITARA (**S**ecure **I**n-camera **T**otal **A**nonymization, **R**estoration on **A**uthorization), a Raspberry Pi–based smart glasses prototype, generalizable to broader wearable and edge camera platforms. This paper makes the following contributions:

- We design a novel opt-in privacy-enhancing obfuscation protocol that blurs bystander faces by default and enforces cryptographic consent for any de-anonymization, ensuring zero raw exposure of recorded media. Our design also ensures strong privacy guarantees against malicious wearers.
- We implement SITARA alongside an edge-based evaluation of its accuracy and system cost, demonstrating the feasibility of on-device privacy protection, mobile-based synthetic face replacement, and a functional cloud-based consent mediator.
- We introduce a lightweight public-key consent protocol that splits decryption privileges between device and cloud, ensuring that no single party can reveal a face without the bystander's digital signature.
- We contribute a self-generated, manually annotated dataset of 16,500 frames captured with Ray-Ban Meta glasses, with face bounding boxes across diverse real-world scenarios.

To our knowledge, SITARA is the first protocol for camera glasses that combines default on-device obfuscation with consent-based restoration. Our Tier-1 blurring pipeline achieves Average Precision (AP) 0.94 and Average Recall (AR) 0.95, ensuring bystanders are reliably anonymized across motion, distance, and multi-person videos. Tier-2 synthetic replacement demonstrates that high-quality visuals remain possible even from blurred inputs, reaching Fréchet Inception Distance (FID) 63.7, Structural Similarity Index (SSIM) 0.61, and Peak Signal-to-Noise Ratio (PSNR) 12.9 dB, competitive with swaps on unblurred faces. Crucially, these protections come with the necessary cost of privacy preservation: the system operates at 2.8× baseline energy use and a modest 23% storage overhead. Furthermore, our protocol incurs a minimal 24% overhead when no bystanders are present, ensuring effective resource utilization. These results demonstrate that our harder-by-design task matches state-of-the-art baselines, establishing the feasibility of privacy-by-default smart glasses. The complete source code for each tier, along with our custom dataset, is available at our GitHub repository [14].

II. Threat Model

While a determined attacker could bypass protections by using an unrestricted device, we nonetheless adopt a rigorous threat model that includes such adversaries to ensure systematic privacy guarantees under well-defined conditions [15]. Side-channel attacks and direct firmware compromise are out of scope. We focus on three primary adversaries:

1) **Malicious Wearer:** Controls the smart glasses and companion smartphone. They may jailbreak the phone or app to extract raw frames or identify bystanders, trying to bypass blurring and access the unaltered visual feed. The wearer cannot jailbreak the glasses themselves.
2) **Compromised Cloud Operator:** The trusted third party (TTP) handling consent is assumed honest-but-curious, yet vulnerable to external attack or legal coercion. The TTP may inspect server memory, storage, or traffic to obtain blurred or unblurred media. TTP-wearer collusion is out of scope, and the device prevents the companion app from sharing media with the TTP.
3) **On-Path Network Attacker:** Monitors, intercepts, or modifies traffic between glasses, phone, and cloud. Their goal is to capture encrypted data and attempt to decrypt it offline.

We assume bystanders who wish to receive consent requests are pre-registered with the TTP, maintain a verified profile, and provide a face embedding. Bystanders are motivated to enroll for auditability: to know if, when, and where they were recorded by smart glasses. To prevent fraud (e.g., fake accounts or spoofed embeddings), the TTP enforces liveness checks during enrollment, consistent with biometric standards [16]. Embeddings may also be derived from profile pictures through existing identity providers (e.g., Meta or Google), but additional authentication is required to guard against fake identities [17]. Crucially, embeddings are never generated without explicit knowledge and informed consent. Unregistered or unreachable bystanders remain permanently blurred, ensuring privacy remains protected in all scenarios.

III. System Design

To address the privacy concerns of both wearers and bystanders, we propose a novel on-device privacy-enhancing obfuscation protocol grounded in our design principles. Following the principle of *privacy at the point of capture*, our system reverses the prevailing opt-out model by automatically blurring bystander faces, shifting the burden of protection from individuals to the technology itself. The wearer is then provided with two post-recording choices. *First*, an AI-based synthetic replacement method substitutes blurred regions with photorealistic, non-identifiable faces, improving *recording usability* while preserving anonymity. This allows the wearer to preserve non-identifying social context (e.g., gaze direction, expressions, and conversational dynamics). *Second*, guided by the principle of *bystander-owned disclosure control*, a consent-based restoration scheme allows a trusted third party (TTP) to mediate consent requests for recovering original faces.

To ensure our protocol withstands the threats outlined in our threat model, we further define two key system requirements. First, to preserve user control and defend against a *malicious wearer* or *network adversary*, all blurring must occur on-device, ensuring that original facial data remains inaccessible without explicit cryptographic consent. Second, to mitigate risks from a *compromised cloud operator*, recordings must never be shared externally, and only minimal metadata can be disclosed under strict conditions.

A. Architecture

We design our protocol as a three-tier architecture shown in Figure 1. Tier 1 runs on the glasses, performing on-device face detection and blurring. Tier 2 runs on the companion phone, offering optional synthetic face replacement to improve media quality while preserving anonymity. Tier 3 uses a TTP to mediate consent requests without exposing recorded media.

1) Tier 1 — On-Device Blurring and Processing (Glasses): Tier 1 runs on the smart glasses, securing data at the point of capture. For every recorded frame (image or video), a lightweight face detector identifies bounding boxes around detected faces. Each face region then undergoes three operations.

First, a lightweight landmark detection model extracts geometric keypoints that contain no identity-specific information yet provide sufficient structure to guide synthetic replacement in Tier 2. Second, we use a tracking based approach to assign each face stream (same face across frames) a dedicated cryptographically secure symmetric key. If a bystander leaves and reappears, another key will be generated to maintain multiple streams for a single bystander. For every frame, the cropped face region, its stream identifier, frame index, bounding box coordinates, and detector confidence are packaged into a face packet. This packet is encrypted with the symmetric key. Finally, a convex hull (outline around landmarks) is computed and blurring is applied, limiting video degradation and preserving compatibility with downstream synthetic replacement.

After frame-level processing, a face embedding model produces a single embedding for each bystander stream, using the detector confidence to select the best-aligned face packet. Selecting the highest-confidence face packets reduces the risk of misidentification by avoiding poor viewing angles or motion blur. These embedding(s) are encrypted with the corresponding

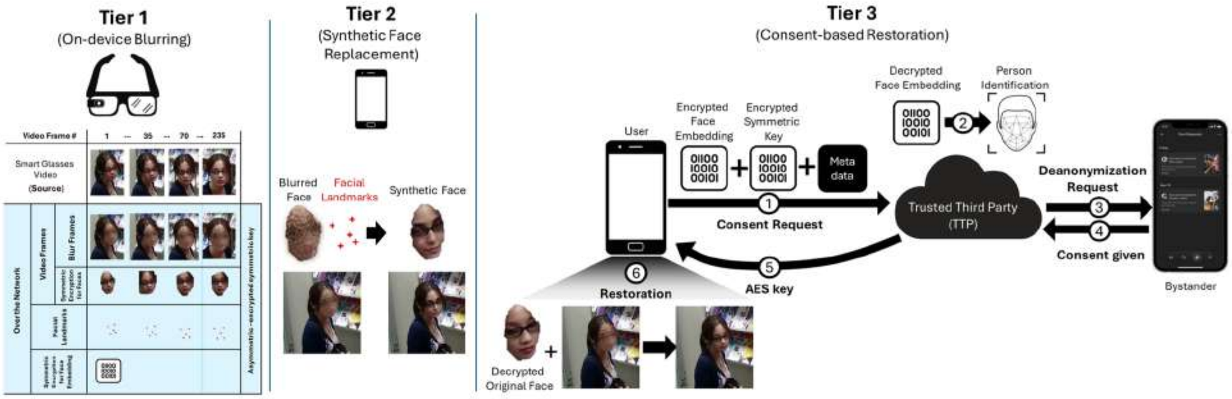

Fig. 1. **Three-tier pipeline for privacy-preserving smart glasses.** Tier 1 (on the glasses) performs on-device face detection, encryption, and blurring. Tier 2 and Tier 3 act as post-recording choices for the wearer. Tier 2 (companion phone) optionally replaces blurred faces with synthetic ones and buffers encrypted face data. Tier 3 (cloud TTP) mediates consent by identifying bystanders via encrypted face embeddings and providing decryption keys upon verified consent.

symmetric key(s). The symmetric key(s) are then encrypted with the TTP's public key, ensuring that only the TTP can manage decryption and that the wearer cannot access unblurred face regions. To enable key rotation and revocation, the smart glasses acquire and maintain the TTP's public key via signed updates from the companion device, which are verified against an embedded manufacturer-backed root key to ensure authenticity. The encrypted keys, face packets, blurred media, and unencrypted landmark points are then transferred securely to the wearer's companion phone via Bluetooth. At this point, the wearer has access to the blurred recording, but lacks the decrypted symmetric keys required to reveal original faces, ensuring strong privacy guarantees at capture.

2) Tier 2 — Landmark-driven Synthetic Replacement (Companion Phone): Tier 2 executes on the companion phone, which offers significantly greater computational capacity than the Tier 1 smart glasses. Since bystander faces are blurred for privacy preservation, conventional face-swapping techniques cannot be applied directly. To address this, we use a landmark-driven, mobile-optimized face replacement method that overlays synthetic identities onto blurred targets.

The replacement process uses the landmarks obtained in Tier 1 to construct a triangular mesh via Delaunay triangulation, allowing the source face to be aligned with the target geometry. The warped result is blended into the blurred region. Finally, a mobile-optimized face-swapping model refines textures, lighting, and background.

3) Tier 3 — Consent-Based Restoration (TTP and Companion Phone): Tier 3 restores the original faces when bystanders grant consent through a TTP. The TTP acts only as a key server and neither accesses nor needs to process recorded media.

The process begins when the companion phone forwards the identifiers of face streams together with their encrypted symmetric keys and encrypted embeddings to the TTP over a secure SSL channel (Figure 1, Tier 3-Step 1). Optionally, the wearer may include limited metadata such as a timestamp, approximate location, or a short personalized message to provide additional context for the request, if they are comfortable sharing this information with the bystander or the TTP. While the system can cryptographically guarantee the validity of the metadata, the free-form personalized message is not inherently trustworthy and should be treated as optional context.

Upon receiving the consent request, the TTP decrypts the symmetric keys with its private key and uses them to access the corresponding embeddings. When multiple embeddings originate from the same bystander, such as when a person leaves and re-enters the frame, the TTP resolves them by computing cosine similarity and merging highly similar streams. The embeddings are then compared against a reference database to identify the bystander (Figure 1, Tier 3-Step 2). If a likely match is found, the TTP issues a consent request through the bystander's registered profile (Figure 1, Tier 3-Step 3).

The matching process must account for the inherent risk of misidentification, which could result in unauthorized data access or "consent fatigue" from incorrectly targeted requests. To mitigate this, we employ a conservative, pre-defined similarity threshold. For our proof of concept, we used cosine similarity with a 0.65 threshold based on initial testing on a database of 15 face embeddings. Future systems should incorporate additional safeguards like aggregating embeddings over multiple frames to reduce the impact of transient noise or poor viewing angles. While these technical safeguards improve system integrity, a full analysis of the legal implications and regulatory compliance is beyond the scope of this work.

If the bystander approves the request (Figure 1, Tier 3-Step 4), the TTP securely transmits the corresponding decrypted symmetric key(s) to the companion phone (Figure 1, Tier 3-Step 5). In the case of repeated appearances in a video, all relevant keys for that bystander are provided. These key(s) are only valid for the bystander(s) who provided consent and cannot be used to decrypt face regions for other bystanders. The companion phone then decrypts the relevant regions and merges them into the blurred media, restoring the original face (Figure 1, Tier 3-Step 6). The entire decryption and restoration process occurs locally on the wearer's phone, and no media is sent to the cloud for obfuscation or restoration.

TABLE I

HARDWARE SPECIFICATIONS: AR1 GEN 1 VS RASPBERRY PI 4 MODEL B.

Specification	AR1 Gen 1	Raspberry Pi 4 Model B
CPU	4 cores @ **1.9 GHz**	4 cores @ **1.8 GHz**
Architecture	64-bit	64-bit
GPU	Qualcomm Adreno	VideoCore VI
NPU	**Yes**	**No**
Tensor accelerators	**Yes**	**No**
Memory Support	LPDDR4	LPDDR4

TABLE II

COMPARISON OF FACE DETECTION MODELS ON WIDER FACE DATASET, SHOWING MODEL SIZE (PARAMETERS), ACCURACY, AND COMPUTATIONAL COST (FLOPS).

Model	Params	FLOPs	Accuracy (Easy/Med/Hard)
SCRFD-2.5GF [20]	0.67M	2.53G	93.78/92.16/77.87
SCRFD-0.5GF [20]	0.57M	0.779G	90.7/88.2/68.4
FDLite [22]	0.24M	0.94G	92.30/89.90/82.30
CenterFace [23]	1.83M	2.06G	**93.20/92.10/87.30**
YuNet [20]	**0.075M**	**0.595G**	**89.90/86.90/69.10**

TABLE III

LANDMARK DETECTION ON 300W DATASET [26]: MODEL SIZE, NORMALISED MEAN ERROR (%), FLOPS, AND LANDMARK COUNT.

Model	Model Size	FLOPs	NME Loss	Landmarks
PLFD-1.0X [28]	12.5MB	300M	3.95	68
HRNetV2 [29]	36.85MB	668.54M	5.37	98
PIPNet [30]	3.01MB	400M	**3.36**	68
Mediapipe*	**2.3MB**	**230.74M**	**10.5**	**468**

Note: Mediapipe FaceMesh NME was calculated using our own experiments.

TABLE IV

COMPARISON OF FACE EMBEDDING MODELS ON LFW DATASET, SHOWING MODEL SIZE (PARAMETERS), ACCURACY AND COMPUTATIONAL COST (FLOPS).

Model	Params	Accuracy (LFW)	FLOPs
GhostFaceNets [34]	6.9M	0.9968	0.2721G
MobileFaceNets [35]	0.99M	0.9955	0.22G
PocketNet [36]	**0.92M**	0.9955	0.6G
EdgeFace [31]	**1.77M**	**0.9973**	**0.197G**
FaceNet [37]	140M	0.9963	1.6G

IV. SITARA

We implement our protocol in a working prototype, SITARA, which instantiates the three-tier architecture to demonstrate feasibility on resource-constrained hardware.

A. *SITARA Tier 1*

We implement Tier 1 of SITARA on a Raspberry Pi 4 Model B [18], using it as a stand-in for the Snapdragon AR1 Gen 1 [19] smart glasses platform that powers the current Meta Ray-Ban collection [1]. As summarized in Table I, the Raspberry Pi has a slightly slower CPU, lacks dedicated machine learning accelerators such as NPUs or tensor cores, and its VideoCore VI GPU is limited to basic video processing and lightweight 3D rendering. Using hardware less capable than today's commercial devices, we validate that SITARA is viable across various platforms, ranging from current smart glasses with ML accelerators to older, resource-limited systems.

We implement the tier 1 pipeline with three threads: a read thread loads frames into a queue, a processing thread handles them and passes results to a write queue, and a write thread stores them. This prevents the processing delays from blocking frame reading and preserves natural FPS. Read and processing threads run simultaneously until video capture ends, after which the processing thread continues. Figure 2 shows power consumption across these stages with a 0.2 s delay between frame processing to illustrate the pattern.

SITARA adopts the Yunet face detector [20] as its primary model. Table II compares Yunet with other state-of-the-art detectors on the Wider Face dataset [21]. As shown in Figure 4, detector inference incurs higher power consumption and introduces substantial latency overhead compared to frame reading alone. To reduce the cost of repeated full-frame detection, SITARA employs a detector-skip strategy (Fig. 3): the detector runs once every 7 frames, or earlier if a motion-based heuristic (frame-difference mean square error) signals a significant scene change. When the detector is skipped, sparse optical flow tracking is used to update bounding boxes.

The trade-off between accuracy and efficiency is shown in Fig. 5, which reports average precision and recall at an IoU threshold of 0.5 (following [24]) across different skip values. A skip value of 7 was selected as the best compromise between detection reliability and processing time on our custom dataset.

For each detected bounding box, SITARA applies the Mediapipe face mesh model [25] to extract a 468-point mapping of facial features. Table III presents its performance relative to other landmark models on the 300W dataset [26]. The convex hull of these landmarks is anonymized using a solid-color blur, while the original face stream is encrypted with the bystander's AES-128 GCM symmetric key [27].

After frame-level processing, SITARA computes a single 512-dimensional embedding for each bystander using the EdgeFace s-gamma-05 embedding model [31]. Table IV compares EdgeFace with alternative embedding models on the Labeled Faces in the Wild (LFW) dataset [32]. Each embedding is encrypted with the corresponding AES-128 symmetric key. The AES key is further encrypted using the TTP's RSA-4096 public key [33], ensuring the wearer cannot use the AES key to decrypt face crops or embeddings.

Finally, the encrypted face streams, embeddings, and AES keys, together with the blurred media and unencrypted landmark points, are securely transmitted to the wearer's companion phone over a secure Bluetooth channel. At this stage, the wearer can view recordings with all bystander faces blurred but cannot access the keys to recover the originals.

B. *SITARA Tier 2*

Tier 2 performs synthetic face replacement on the companion phone using the blurred faces along with their 468-point landmark mapping. For each face stream, a synthetic identity is selected from a curated database of AI-generated faces, each pre-annotated with the same set of landmarks. Selection is guided by approximate skin tone matching between the blurred target and candidate synthetic faces. To address deepfake concerns, synthetic faces are generated from a fixed pool, and the companion app prevents users from adding arbitrary faces.

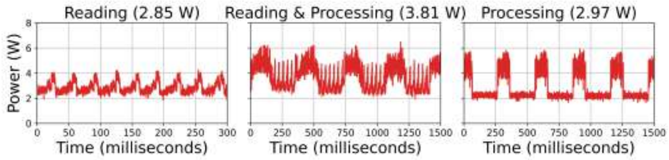

Fig. 2. Average power consumption across protocol stages.

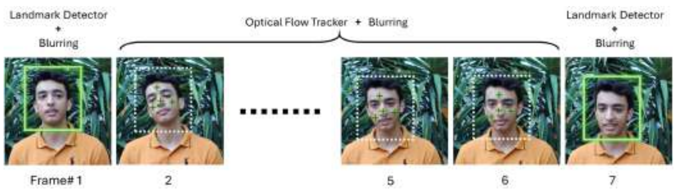

Fig. 3. Detector-skip with optical-flow tracking.

Once a source face is chosen, SITARA applies a geometric warping pipeline to align it with the target. Three critical landmarks, the nose tip and mouth corners, are used to compute an affine transformation, through the Least-Median robust method to improve facial alignment. The transformed face is integrated into the blurred region using Poisson image editing, with the convex hull of the target landmarks defining the blending area. To ensure temporal consistency, the pipeline also maintains a per-face landmark history across frames.

Since conventional warping techniques produce significant visual artifacts such as discontinuities and misaligned features, the warped face is refined using MobileFaceSwap [38], a mobile-optimized face-swapping model that seamlessly blends textures, lighting, and background consistency. The complete synthetic replacement pipeline is deployed as an Android application, enabling efficient execution on mobile hardware and validating the practicality of Tier 2 in SITARA.

C. SITARA *Tier 3*

Tier 3 implements consent-based restoration by simulating the role of a TTP[1] on a remote server equipped with a prototype database of 15 bystanders. When a wearer initiates a consent request, the companion phone transmits the consent file and selected metadata to the TTP over a secure SSL channel. The TTP then compares the encrypted embeddings against its database using cosine similarity to identify potential matches. If a bystander grants consent, the TTP releases the corresponding decrypted AES-128 key(s) to the companion phone through the same secure channel.

The companion phone uses the received keys to decrypt the relevant face regions and seamlessly merge them back into the blurred media. All decryption and restoration occurs locally on the wearer's phone, ensuring that raw media is never uploaded to the cloud. We implement the decryption and restoration mechanism through a companion Android application using the OpenCV Android SDK.

[1]Aside from a specialized privacy platform, the TTP could be implemented by platforms such as Facebook, which already possess large datasets of profiles and provide highly reliable identity verification. The precise nature and deployment of the TTP, however, lies beyond the scope of this paper.

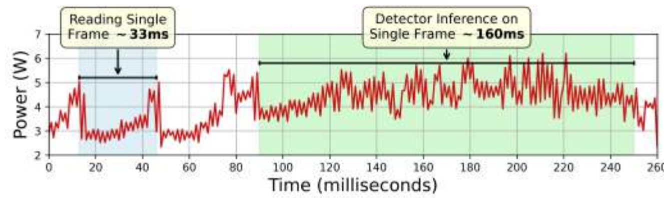

Fig. 4. Comparing frame reading & face detector inference (RPi 4B)

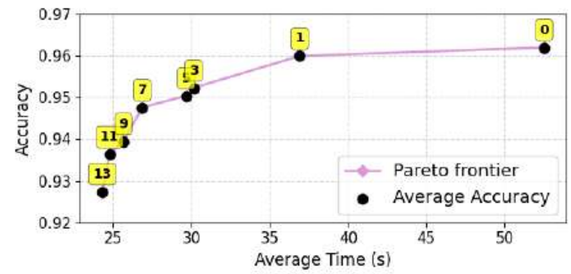

Fig. 5. **Accuracy vs. Processing Time (Skip Values):** Skip 7 lies on the Pareto frontier. Beyond Skip 7, AP/AR declines with marginal time gains.

V. Evaluation

We evaluate SITARA across four dimensions: *privacy protection*, *visual utility*, *system cost*, and *user perceptions*.

1) **Privacy protection** quantifies the effectiveness of our Tier 1 blurring pipeline. We report Average Recall (coverage of ground-truth faces) and Average Precision (avoiding false or excessive blurring), using an IoU threshold of 0.5 under the MS-COCO protocol [24].
2) **Visual utility** measures the perceptual quality of synthetic replacement using standard metrics: Fréchet Inception Distance (FID), Structural Similarity (SSIM), Peak Signal-to-Noise Ratio (PSNR), Learned Perceptual Image Patch Similarity (LPIPS), and a 468-point landmark error.
3) **System cost** captures processing latency, energy consumption, and storage overhead on wearable-class hardware.
4) **User perceptions** qualitatively analyzes how camera glass wearers and bystanders perceive our Tier 3 consent-based restoration mechanism.

We benchmark performance on a custom dataset captured with Ray-Ban Meta Stories glasses in varied real-world settings. Clips are categorized by bystander count (1–5), face size (far, medium, close), and motion (static, bystander moving, wearer moving), as illustrated in Figure 6. The dataset comprises 16,500 annotated frames with manually labeled face bounding boxes, and we release both videos and annotations for reproducibility. For contrast, we also report results on the `CCV2` dataset [39], which consists of single-subject recordings. Together both datasets cover an extensive range of scenarios, however they do not include partially occluded faces. Future work should carefully examine the impact of such cases on detection and matching accuracy. We focus the evaluation

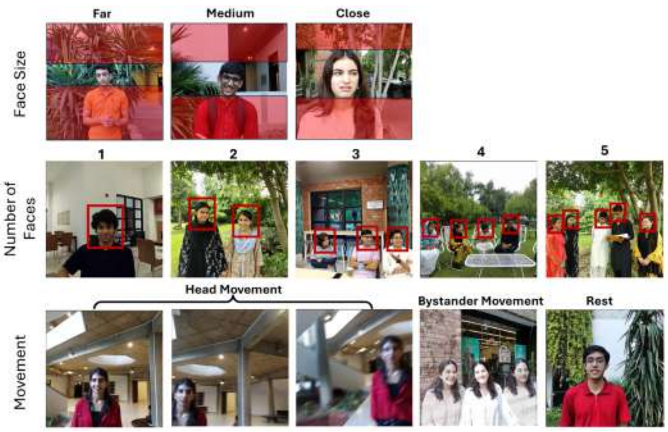

Fig. 6. Dataset overview showing variations in face size, number of faces, and movement.

section on the resource-constrained edge device (Tier 1/2) because Tier 3 latency is dominated by standard network RTT.

A. Privacy Protection: Obfuscation Effectiveness

We evaluate the Tier 1 blurring pipeline using both datasets. From each, we sample 16,500 frames (5 frames per second) and manually annotate face bounding boxes. Four independent annotators conducted labeling using `LabelImg` [40]. As a baseline, we compare against *EgoBlur* [12], a state-of-the-art blurring system. For EgoBlur, bounding boxes are taken directly from its detector output. For SITARA, convex hulls are computed from detected landmarks, bounding boxes are drawn around them, and blurring is applied. Predictions are then compared with ground-truth annotations.

Metrics: We report AP and AR using IoU 0.5, following the MS-COCO protocol [24] and prior work [12]. Detector confidence is kept at 0.5.

Overall Accuracy: Table V shows results on both datasets. On the simpler CCV2 clips, SITARA achieves AP 0.98 and AR 0.99, nearly identical to EgoBlur (0.99, 0.99), demonstrating parity in controlled settings. On our real-world dataset, SITARA attains AP 0.9421 and AR 0.9531, comparable to EgoBlur (0.9354, 0.9736). Notably, SITARA improves precision (94.21% vs. 93.54%) with a modest recall trade-off (95.31% vs. 97.36%). These results confirm that enforcing blurring at capture time preserves detection quality, delivering privacy protection on par with server-based solutions while operating on edge hardware.

Performance Under Varying Conditions: Table VI summarizes results by movement state, face count, and face size. Accuracy for both models decreases as the number of faces increases, with strong performance up to three faces but showing sharper declines at four (AP 0.900) and five (AP 0.882). For face size, SITARA outperforms EgoBlur on far (AP 1.000 vs. 0.930) and medium (AP 0.990 vs. 0.856) ranges, where detection is most difficult, while both pipelines perform similarly on close-range faces. These results highlight SITARA's strength in long-range scenarios where bystanders appear incidentally in the background.

TABLE V

SITARA Tier 1 and EgoBlur Performance Comparison.

Method	CCV2 AP	CCV2 AR	Custom Dataset AP	Custom Dataset AR
Our Pipeline	0.98	**0.99**	**0.9421**	0.9531
EgoBlur	**0.99**	**0.99**	0.9354	**0.9736**

TABLE VI

AP/AR performance by face count, mobility, and scale. Best results per row are highlighted.

Category	Sub-Category	Our Pipeline AP	Our Pipeline AR	EgoBlur AP	EgoBlur AR
Number of Faces	One Face	**0.990**	0.997	0.932	**1.000**
	Two Face	**0.967**	0.979	0.963	**0.982**
	Three Face	**0.979**	**0.982**	0.974	0.981
	Four Face	0.900	0.904	**0.909**	**0.934**
	Five Face	0.882	0.917	**0.952**	**0.969**
Movement State	Rest	**0.990**	0.998	0.971	**0.999**
	Head	0.920	0.947	**0.925**	**0.980**
	Bystander	0.959	0.968	**0.961**	**0.983**
Face Size Range	Far	**1.000**	**1.000**	0.930	**1.000**
	Medium	**0.990**	0.992	0.856	**0.993**
	Close	**0.990**	0.997	0.989	**1.000**

B. Visual Utility: Synthetic Replacement Quality

Setup & Baseline: In the absence of existing methods for synthetic replacement of blurred faces, we use *MobileFaceSwap (MFS)* [38] as a proxy upper bound on visual quality. MFS operates on unblurred faces, reflecting the best achievable perceptual quality. We focus our evaluation towards establishing watchability without revealing identity, rather than maximizing photorealism. We evaluate all categories of our dataset, excluding variation in face count since results extrapolate from single-face cases. Standardized face crops are sampled for consistency with prior evaluation tiers. For SITARA, blurred crops with 468-point facial landmarks are processed by our pipeline; for MFS, unblurred crops are used. Figure 7 illustrates the input differences and provides qualitative examples of the achieved perceptual quality. We compare both MFS and SITARA outputs against the original unblurred video as the ground truth. In both cases, we standardize crop size and alignment before computing perceptual metrics.

Metrics: We employ widely adopted measures for face generation and swapping. Fréchet Inception Distance (FID) [41] captures distribution-level realism. Structural Similarity Index (SSIM) [42] and Peak Signal-to-Noise Ratio (PSNR) quantify pixel-level fidelity. Learned Perceptual Image Patch Similarity (LPIPS) [43] reflects human perceptual sensitivity. Finally, landmark distance compares the original 468-point landmarks to post-synthetic landmarks to evaluate preservation of geometry and expressions. Importantly, all metrics are agnostic to input condition (blurred vs. unblurred), enabling fair comparison between SITARA and MFS.

Accuracy: Table VII reports the mean and deviation for each metric, with arrows indicating the desired direction. Figure 8 shows the per-category deltas across metrics. Performance is strongest for *far* faces, where limited fine-grained detail reduces artifact visibility and narrows the gap with MFS. *Close*

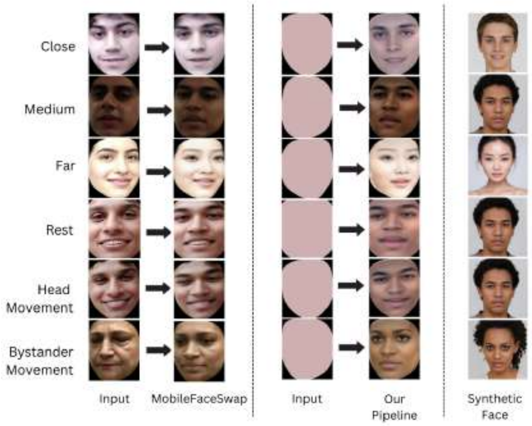

Fig. 7. Synthetic face replacement evaluation for baseline and SITARA.

TABLE VII
METRIC EVALUATION (AVG $\pm$ STD). ARROWS ($\uparrow/\downarrow$) INDICATE PREFERRED DIRECTION. LAST COLUMN SHOWS THE THEORETICAL RANGE.

Metric	Baseline	Our Pipeline (Sitara)	Range
FID $\downarrow$	31.00 ± 13.83	63.70 ± 27.78	≥ 0 (no upper bound)
SSIM $\uparrow$	0.76 ± 0.07	0.61 ± 0.07	$[0, 1]$
PSNR (dB) $\uparrow$	15.87 ± 2.77	12.85 ± 1.95	$[0, \infty)$ dB
LPIPS $\downarrow$	0.14 ± 0.06	0.27 ± 0.07	$[0, 1]$
Landmark Distance $\downarrow$	8.81 ± 3.99	15.94 ± 7.13	$[0, \infty)$

and *medium* faces preserve higher-frequency detail, making differences more noticeable and producing larger deltas. Under *bystander movement*, SITARA remains robust with only modest SSIM and PSNR degradation from background-induced inconsistencies. *Head movement* is most challenging, as rapid pose changes increase alignment difficulty and produce the largest deltas. In summary, despite operating on blurred faces SITARA preserves visual utility close to state-of-the-art face swaps on unblurred inputs, showing that privacy by default does not compromise the wearer's experience.

C. System Cost

We evaluate SITARA's system overhead to verify feasibility on wearable hardware, focusing on Tier 1 (on-glasses blurring and encryption) as the critical component.

1) Performance Efficiency: We assess Tier 1 performance on prototype hardware using two metrics: (i) end-to-end processing time and (ii) energy consumption for battery impact. Five 10-second clips were recorded per dataset category with a frame-skip value of seven. As a baseline, we recorded the video with the protocol disabled to capture idle processing and I/O overhead. Each experiment was repeated five times under identical conditions. Frame storage was excluded since it is constant for both baseline and our pipeline.

Measurement Setup: Experiments used a Raspberry Pi 4 Model B [18] with a Pi Camera module V1.3, chosen for its similarity to smart-glasses cameras. Encrypted metadata was redirected to a USB flash drive since smart glasses come with flash capabilities. The selected USB drive (15 MB/s peak) approximates the 12–45 MB/s write speed of e.MMC flash used in Meta glasses [44]–[46]. The Pi ran a 64-bit headless OS with background processes disabled. Power was supplied via 5 V and GND header pins from a regulated source. Current was measured with a 0.1 Ω precision shunt resistor and oscilloscope probes. Instantaneous power was calculated as $P = VI$. Noise was separately characterized and subtracted. Experiments were conducted at room temperature ($\sim 25°$C). Figure 9 shows the measurement setup.

Results: Since the landmark model is needed only for synthetic face replacement, we separate costs between the core obfuscation-restoration process and the extra overhead of replacement. Across all categories, SITARA required 22.88 s for the full protocol and 13.69 s for the obfuscation–restoration mechanism alone, compared to a 10 s baseline. Average energy consumption was 112.05 J for the full protocol and 67.04 J for the core mechanism, against a 40.0 J baseline. Figures 10 and 11 show per-category breakdowns for latency and energy consumption. The protocol performs best on videos without faces, showing zero latency and minimal energy use, proving its efficiency in non-social contexts. Within the face-size category, smaller faces lower tracking and encryption costs, improving energy and latency. Overheads rise with more faces as landmark inference scales linearly, though the core privacy mechanism remains manageable. Figure 13 shows one power trace per category, detailing the pipeline's energy behavior.

Impact on Battery Life: Since the Raspberry Pi is a general-purpose platform rather than an energy-optimized device, we estimate energy use via ratios. Most manufacturers, including Meta, do not report battery life under continuous recording, so we use the reported 50-minute continuous recording baseline from an open-source smart glasses platform [47]. Compared to our 40.0 J energy baseline, total pipeline energy is 2.8× higher, while the core privacy mechanism alone is only 1.68× higher, yielding approximate continuous recording times of 17.9 and 29.8 minutes, respectively. While these costs seem high under constant use, they mainly occur when bystanders are present and reflect the cost of strong privacy guarantees. For typical intermittent use, the impact on battery life is much smaller.

2) Storage Overhead: SITARA requires on-device storage to save encrypted face regions and landmarks for Tier 2 replacement. We compare this overhead against the baseline cost of storing video alone. Across our custom dataset, SITARA adds an average of 14.06 MB per 30-second clip, relative to a baseline size of 56.16 MB. Figure 12 shows the breakdown across categories. Overhead is highest in close-range videos, where larger face regions produce bigger encrypted blocks. Interestingly, storage overhead does not increase with the number of faces, even though more encrypted crops must be stored. This is because our dataset captures uncontrolled, real-world scenarios: videos with more faces typically have subjects at a greater distance, resulting in smaller face sizes. This demonstrates that storage overhead scales very efficiently in real-world, multi-person settings. Moreover, storage overhead drops to zero when no bystanders are present, highlighting the system's modular design. Most Meta smart glasses ship with 32 GB of onboard flash, sufficient for 500+ photos or 100+ 30-

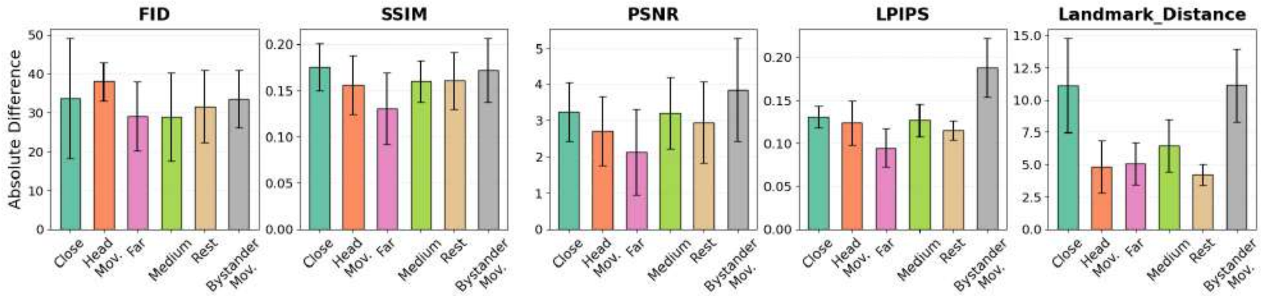

Fig. 8. Absolute mean differences between SITARA and MFS across metrics.

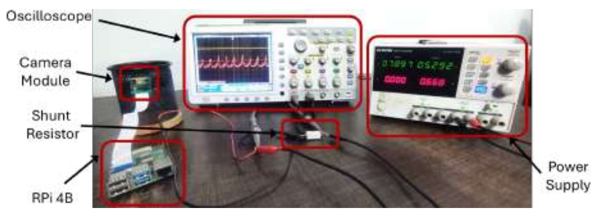

Fig. 9. Energy measurement setup.

Fig. 10. Average total processing time across categories on RPi 4 Model B.

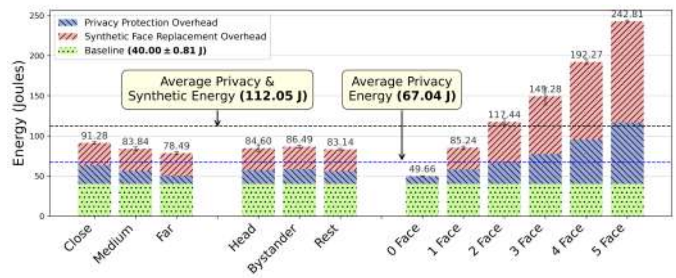

Fig. 11. Average energy consumption across categories on RPi 4 Model B.

second videos [1]. Against this baseline, SITARA introduces an additional 23% storage overhead, reducing the effective capacity to approximately 400+ photos or 80+ videos.

D. User Perceptions

To understand how users perceive the Tier 3 consent mechanism, we share preliminary results from a user study ($N = 18$) with camera glass wearers ($W = 9$) and bystanders ($B = 9$) [48]. Participants interacted with the protocol interface and discussed their perceptions in semi-structured interviews.

Our findings reveal that bystanders strongly supported mandatory opt-in obfuscation mechanisms. While bystanders reported being highly likely to accept consent requests from friends and family, they required more context from strangers, such as the intended use of the recording. Wearers found certain protocol features, such as the default blurring state, to be cumbersome in social contexts with friends or family. However, participants were receptive to a more context-dependent approach, such as enforcing the protocol in unknown environments while relaxing it in familiar places or with pre-approved individuals.

Summary: Overall, SITARA balances privacy, utility, and efficiency. Tier 1 blurring enforces strong bystander protection on edge devices. Tier 2 synthetic replacement maintains usability with outputs comparable to face swaps on unblurred footage. Our resource efficient design ensures that overheads scale only when bystanders are present. These costs remain within modern device limits and represent the necessary trade-off for privacy at capture. Our preliminary qualitative results show how users perceive the protocol in real-world scenarios. Overall our results demonstrate that privacy-by-default smart glasses are feasible: bystanders can be reliably anonymized while still enabling high-quality, usable recordings.

VI. Related Work

Prior research on PETs for cameras has laid important foundations, but none fully resolve the tension between strong privacy and post-capture usability. Our work builds on these efforts while introducing a consent-driven, reversible approach tailored to wearable devices.

A. Existing Privacy-Enhancing Technologies (PETs)

Early PETs placed the burden on bystanders, requiring them to broadcast do-not-capture signals (e.g., FaceBlock [8], I-Pic [11], and wearable mediators [49]). Such systems provided only conditional privacy and permanently discarded data. A second generation, including EgoBlur [12] and Google Street View, enforced privacy-by-default by automatically blurring faces. While stronger, these systems irreversibly destroyed visual information, limiting utility even when consent could be granted. SITARA balances this trade-off by combining privacy-by-default, cryptographic consent, and synthetic replacement to recover visual data with bystander authorization.

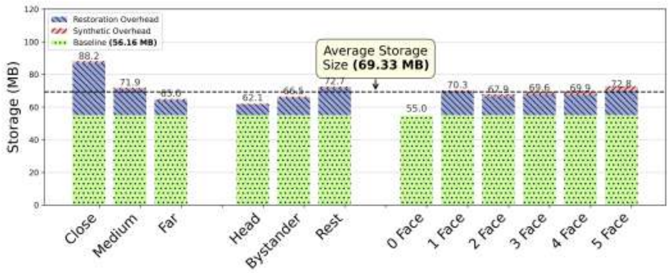

Fig. 12. Average storage overhead across categories. The encrypted face region overhead is shown in blue, overhead for landmarks is shown in red.

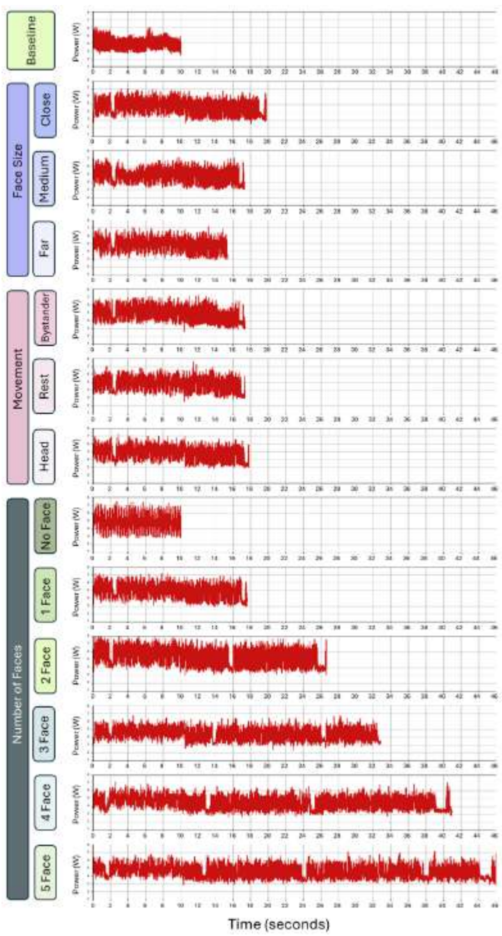

Fig. 13. Power consumption traces for each category on RPi 4 Model B.

B. Synthetic Face Anonymization

Generative approaches (e.g., DeepPrivacy [50]) have been extensively used for de-identification while preserving context, but they are typically irreversible. Recent efforts at reversible anonymization, such as password- or key-based restoration [51], [52], vest control in the operator holding the secret rather than the bystander. In contrast, SITARA enforces bystander-governed restoration: per-face keys are cryptographically bound to bystander consent, ensuring neither wearer nor cloud can unilaterally reveal identities.

C. Subject vs. Bystander Privacy

A fundamental distinction for Privacy-Enhancing Technologies (PETs) is determining whose privacy requires protection. PETs such as BystandAR suggest that only the privacy of background bystanders is relevant, assuming that primary subjects are likely aware of the recording and provide implicit consent [53]. Such systems often rely on error-prone heuristics like eye-gaze to differentiate between subjects and bystanders, where unintentional (or malicious) gaze can accidentally classify a bystander as a subject. Early PETs, such as I-Pic, allowed individuals to broadcast privacy preferences by distinguishing between ”friends” (subjects) and ”strangers” (bystanders) [11]. While subject-focused systems were primarily designed for head-worn AR devices with explicit form factors that signaled recording, modern camera glasses have become increasingly indistinguishable from regular eyewear. Although manufacturers include LED and audio indicators to address these concerns, research has proven them ineffective. Portnoff et al. [54] demonstrated that fewer than 50% of users noticed LEDs during computer tasks, and Koelle et al. [55] criticized such indicators for poor visibility and clarity. Bhardwaj and Ponticello et al. [7] further confirmed that both audio and visual indicators fail to reliably notify subjects.

To tackle these concerns, we adopt a ”Zero-Trust” approach by encrypting all captured individuals by default. Under this model, a subject can easily grant consent later via Tier 3, whereas a bystander cannot ”un-reveal” themselves once their privacy has been compromised. Regardless, our modular design ensures the system can be adapted for head-worn devices by integrating a subject vs. bystander classifier [56].

VII. Conclusion

Smart glasses highlight a critical tension between utility and privacy. SITARA resolves this by enforcing privacy at capture while enabling consent-driven recovery and preserving usability through synthetic replacement. Our three-tier design achieves strong guarantees: Tier 1 blurring anonymizes bystanders with AP 0.94 and AR 0.95, matching or surpassing state-of-the-art baselines even under motion and crowded scenes. Tier 2 demonstrates that synthetic replacement can deliver perceptual quality competitive with face swaps on unblurred inputs, showing that privacy need not degrade the wearer’s experience. Tier 3 introduces a novel cryptographic consent protocol, ensuring that restoration is possible only with explicit bystander authorization. Despite the harder-by-design setting, SITARA remains feasible on wearable-class hardware with manageable energy (2.8× baseline) and minimal storage (23%) overheads, proving deployability. Our design further ensures that these overheads represent only the necessary cost of preserving privacy, dropping to a minimal 24% energy increase when no bystanders are present. These results establish the feasibility of privacy-by-default smart glasses that balance the needs of both wearers and bystanders, offering a path toward responsible AR adoption.

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